

Towards fast and constant-time implementations of Hamming Quasi-Cyclic

A case study on ARM Cortex-M4 devices

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Overview

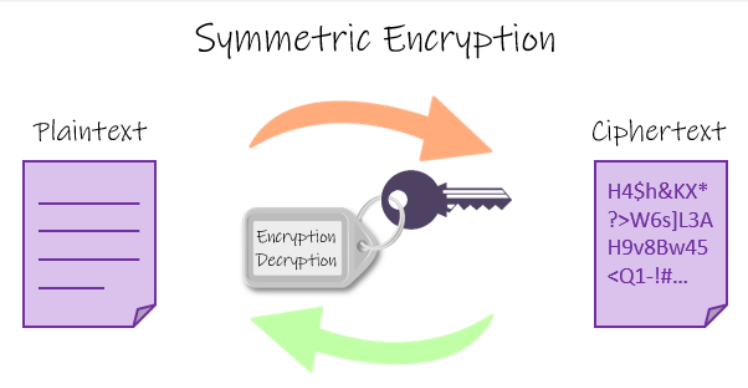
1. Introduction
2. Background
3. Implementation
4. Experimental results
5. Conclusions

Introduction

Kerchoff's Principle

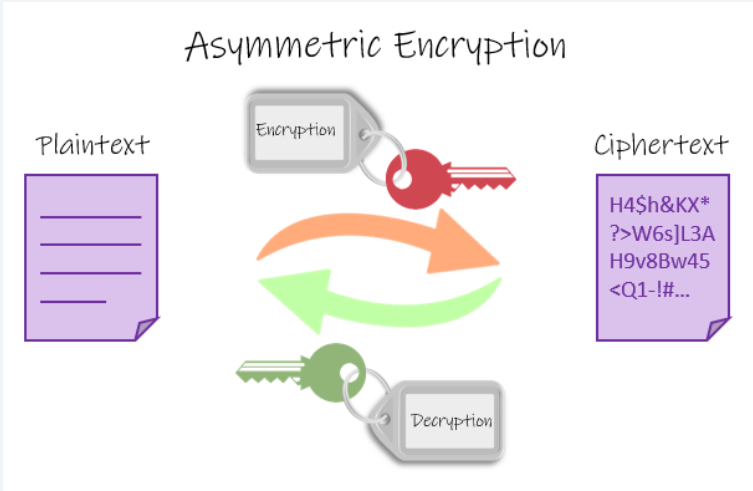
Everything about a cryptosystem is publicly known, **except for the key**

Symmetric encryption



Key: a n-bit string

Asymmetric encryption



Key: the product of two prime numbers, *i.e.* $n = p \cdot q$

Cryptanalytic attacks

What are the best attacks on modern cryptosystems?

On Classical Computers

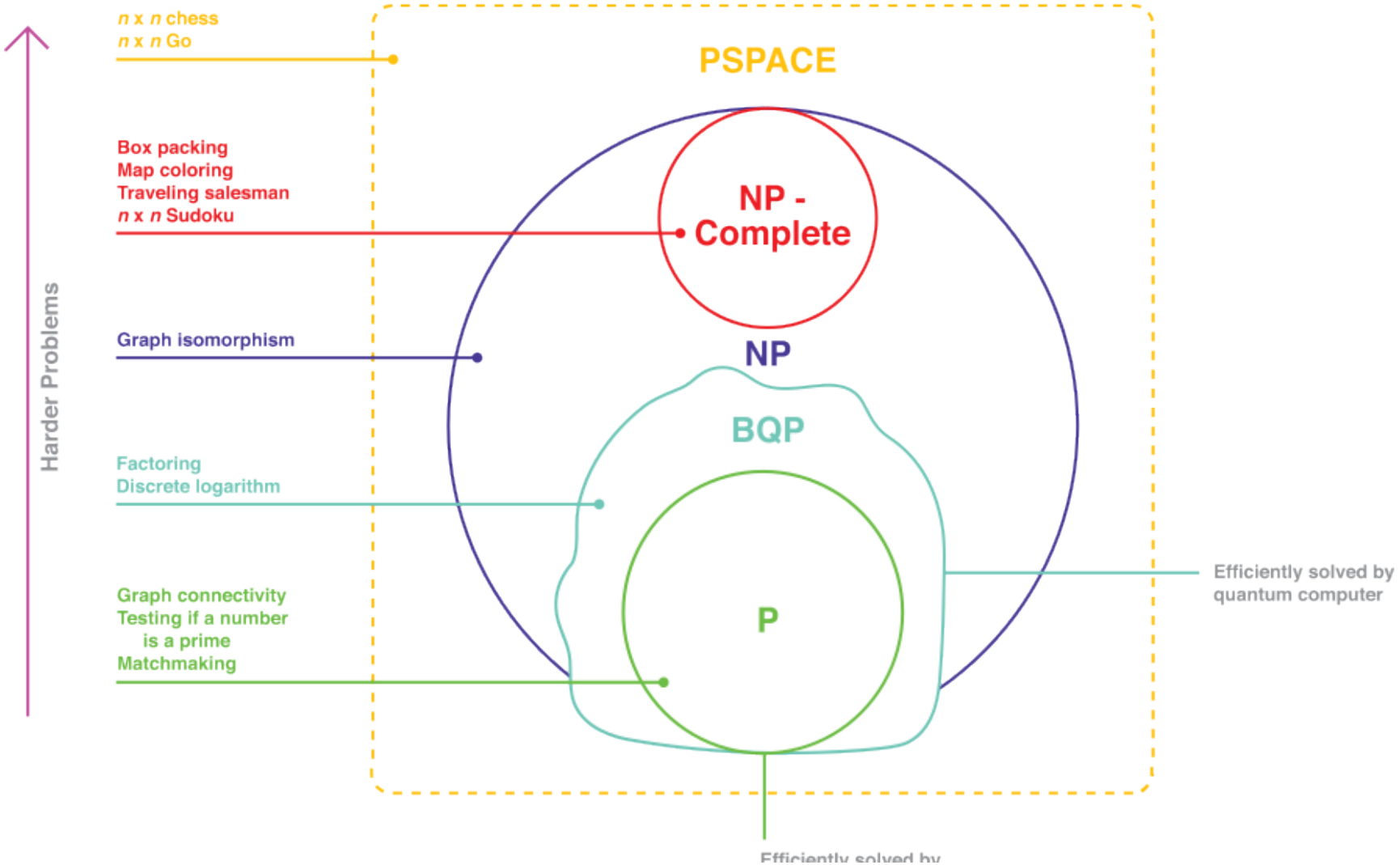
- Symmetric: exhaustive search, $\mathcal{O}(2^n)$
- Asymmetric: General Number Field Sieve, $\approx \mathcal{O}(2^{n-\epsilon})$

On Quantum Computers

- Symmetric: Grover's algorithm, $\mathcal{O}(\sqrt{2^n})$
- Asymmetric: Shor's algorithm, $\approx \mathcal{O}((\log n)^2)$

Shor's algorithm can break asymmetric cryptosystems in polynomial time!

The Computational Complexity Zoo



NIST Post-Quantum Cryptography Standardization

Timeline

- 26/02/16: Announcement of NIST's call for submissions
- 21/12/17: Announcement of the first round candidates (69)
- 30/01/19: Announcement of the second round candidates (26)
- 22/07/20: Announcement of the third round candidates (7 finalists, 8 alternatives)
- 05/07/22: Announcement of the fourth round candidates (4)
- 2022-2024: Publication of the standard drafts

Standardization candidates

Third Round winner

- CRYSTALS-Kyber: lattice-based

NIST decided to standardize at least another PQC scheme, in case lattice-based crypto fails

Fourth Round candidates

- ~~**S**upersingular **I**sogeny **K**ey **E**ncapsulation~~
- **B**it Fl**i**pping **K**ey **E**ncapsulation [ABB+22]
- **H**amming **Q**uasi-**C**yclic [MAB+18]
- Classic McEliece [BCC+22]

BIKE, Classic McEliece and HQC are all code-based cryptosystems.

Overview

The target of this thesis is Hamming Quasi-Cyclic

Goals

- High performance implementation for Cortex-M4
- Protection against timing attacks [GHJ+22, WBB+22]
- Protection against power attacks [SRSW20, SHR+22]

Roadmap

- Evaluate various polynomial multiplication algorithms
- Assess the constant-time behavior of all components
- Implement a masking scheme for the PKE primitives
- Compare the performance with the other Round 4 candidates

Background

Implementation

Experimental results

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